

Ebuka Philip Oguchi, Ph.D.

CONTACT INFORMATION

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ACADEMIC APPOINTMENTS

Assistant Professor (Tenure Track) Jan. 2026–Present
Department of Computer and Information Sciences
Spelman College

EDUCATION

University of Nebraska–Lincoln Aug. 2025
Ph.D. in Computer Science
GPA: 4.0/4.0
Advisor: Prof. Nirnimesh Ghose
Dissertation: *Authentication and Message Integrity Verification for Emerging Wireless Networks*

Changchun University of Science and Technology, China 2020
M.S. in Computer Applied Technology
Advisor: Prof. Fang Ming
Thesis: *Smoke Recognition Using ResNet and GoogleNet*

Nnamdi Azikiwe University, Nigeria 2016
B.Eng. in Electronic and Computer Engineering
Senior Project: *Biometric Course Registration System Using Fingerprint Authentication*

RESEARCH INTERESTS

Cybersecurity; Authentication and Message Integrity; Wireless and Physical-Layer Security; Cyber-Physical Systems and Internet of Things; Distributed Confidential Computing; Trustworthy and Secure AI.

PUBLICATIONS

Peer-Reviewed Journal Articles

1. **Oguchi, E.**; Ghose, N.; Vuran, M.C.
Soil-Assisted Trust Establishment for Underground Wireless Networks.
IEEE Transactions on Dependable and Secure Computing (TDSC).
Accepted for Publication, 2026.

2. Anderson, M.*; Duong, T.T.*; **Oguchi, E.***; Wisniewska, A.; Ghose, N.
A Review of Confidentiality, Integrity, and Availability Challenges in Molecular Communications.
 IEEE Transactions on Molecular, Biological, and Multi-Scale Communications.
 Accepted for Publication, 2026.
 *Equal contribution.

Peer-Reviewed Conference Papers

1. **Oguchi, E.**; Lado, H.; Evans, R.; Menon, A.; Wang, B.; Ghose, N.
LocaRadio: RF Fingerprint-Based Location Authentication for Wireless Networks.
 2027 IEEE Consumer Communications and Networking Conference (CCNC), Las Vegas, Nevada, January 2027.
 Accepted for Presentation.
2. **Oguchi, E.**; Ghose, N.; Vuran, M.C.
VET: Autonomous Vehicular Credential Verification Using Trajectory and Motion Vectors.
 EAI SecureComm, Hong Kong, 2023.
3. **Oguchi, E.**; Ghose, N.; Vuran, M.C.
STUN: Secret-Free Trust Establishment for Underground Wireless Networks.
 IEEE INFOCOM Workshops, 2022.

Manuscripts in Preparation

1. *Systematization of Knowledge on Confidential Computing: From Isolated TEEs to Compositional Assurance.*
 Manuscript in preparation with collaborators from the Center for Distributed Confidential Computing.

RESEARCH SUPPORT

- **Collaborator**, NSF Center for Distributed Confidential Computing (CDCC) 2026–Present
 Multi-institutional NSF Frontiers Center supporting research in confidential computing, secure distributed systems, and trustworthy computing.
- **Collaborator**, Google.org Cybersecurity Clinic 2026–Present
 Supporting the establishment of the Spelman College Cybersecurity Clinic and integration of clinic activities into the undergraduate cybersecurity curriculum.
- **Graduate Research Support**
 Supported through NSF awards CNS-2331191, CNS-2225161, CNS-1619285, and ECCS-2030272.

TEACHING EXPERIENCE

Assistant Professor (Tenure Track) Jan. 2026–Present
 Department of Computer and Information Sciences
 Spelman College

Fall 2026

- SCIS 216: Computer Organization and Design (Enrollment: 29)
- SCIS 216L: Computer Organization and Design Laboratory (Enrollment: 29)
- SCIS 123: C++ Programming (Enrollment: 10)
- SCIS 123L: C++ Programming Laboratory (Enrollment: 10)

Spring 2026

- SCIS 216: Computer Organization and Design (Enrollment: 24)
- SCIS 216L: Computer Organization and Design Laboratory (Enrollment: 24)

Spring 2027 (Scheduled)

- SCIS 121: Computer Science I (C++) (For CS and Non-CS Majors)
- SCIS 121L: Computer Science I Laboratory
- SCIS 343L: Operating Systems and Computer Architecture Laboratory

Teaching Assistant

Fall 2022

Cryptography and Security (CSCE 477/877)

University of Nebraska–Lincoln

- Assisted with laboratories covering RSA, AES, hash functions, digital signatures, authentication protocols, and network security.
- Guided students in implementing cryptographic algorithms and analyzing protocol security.

Teaching Assistant

Jan. 2018–May 2018

Computer Architecture and Software Laboratory

D.S. Adegbenro ICT Polytechnic, Nigeria

RESEARCH
EXPERIENCE**Assistant Professor**

2026–Present

Department of Computer and Information Sciences

Spelman College

- Conduct research on authentication and message-integrity verification for emerging wireless and cyber-physical systems.
- Develop security methods that combine wireless measurements, signal processing, and machine learning.
- Mentor undergraduate students in cybersecurity and wireless-systems research.
- Collaborate with researchers in the NSF Center for Distributed Confidential Computing and contribute to the Google.org Cybersecurity Clinic initiative at Spelman College.

Research Assistant

2021–2025

Department of Computer Science and Engineering

University of Nebraska–Lincoln

- Developed the **STUN (Secret-Free Trust Establishment for Underground Wireless Networks)** protocol, a physical-layer authentication framework for secure bootstrapping in Agricultural Internet of Things (Ag-IoT) environments. The work was

published at IEEE INFOCOM Workshops and later extended into an accepted journal article in IEEE Transactions on Dependable and Secure Computing.

- Developed the **VET (Vehicular Credential Verification)** framework for autonomous and connected vehicle networks by integrating trajectory information, motion vectors, Doppler characteristics, and wireless signal measurements to verify transmitter authenticity.
- Investigated RF fingerprinting and location authentication techniques using wireless channel characteristics, signal processing, and machine learning for underground and over-the-air communication environments.
- Conducted research on lightweight security mechanisms for molecular and nano-scale communication networks, including authentication, message integrity, and threat modeling for biological communication systems.
- Designed and implemented wireless experimental testbeds using Software Defined Radios (USRPs), GNU Radio, MATLAB, Python, and embedded hardware platforms.
- Contributed to externally funded NSF research projects involving secure wireless systems, underground communications, cyber-physical systems, and confidential distributed computing.

UNDERGRADUATE RESEARCH MENTORING	Spelman College 2026–Present • Mentor undergraduate students in cybersecurity, wireless systems, and related computing research. • Provide guidance on literature reviews, experimental design, software implementation, data analysis, and technical communication. • Support undergraduate research projects and independent studies.
AWARDS AND HONORS	• Outstanding Ph.D. Dissertation Award, University of Nebraska–Lincoln (2025) • Mary E. and Elmer H. Dohrmann Fellowship, University of Nebraska–Lincoln (2023) • Chinese Government Scholarship (2018) • Outstanding Graduate Student Recognition, University of Nebraska–Lincoln
PROFESSIONAL SERVICE	• Reviewer, IEEE Transactions on Dependable and Secure Computing (2026–Present) • Reviewer, IEEE Transactions on Industrial Informatics • Reviewer, ACM Transactions on Cyber-Physical Systems • Reviewer, ACM WiSec • Reviewer, EAI SecureComm • Judge, Spelman College Hack-A-Bot Competition (2026) • Judge, University of Nebraska Engineering Poster Competition
PROFESSIONAL DEVELOPMENT	• Fellow, ACS Teaching and Learning Workshop, Rollins College (2026) • Faculty Onboarding and Mentorship Program, Spelman College (2026–Present) • CITI Responsible Conduct of Research Training
MEMBERSHIPS	• Institute of Electrical and Electronics Engineers (IEEE) • National Society of Black Engineers (NSBE) • Google Computer Science Research Mentorship Program (CSRMP) • Institute for African-American Mentoring in Computing Sciences (iAAMCS)

TECHNICAL SKILLS **Programming and Analysis:**

Python, C/C++, MATLAB, MIPS Assembly

Wireless and Embedded Systems:

USRP, GNU Radio, Embedded Systems, Breadboard Prototyping

Security and Machine Learning:

Authentication, Physical-Layer Security, RF Fingerprinting, PyTorch, TensorFlow, Scikit-learn

Research Tools:

Linux, Git, LaTeX, Docker